Abhishek Sairam Gaduputi

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SUMMARY

Machine learning engineer with 3+ years of turning models into shipped products, from recommendation systems at a consumer startup to vision-guided autonomous tractors. Now pursuing an MS in AI at NJIT, focused on agentic systems and generative models, and a regular at NYC and NJ hackathons. **Seeking full-time ML or AI Engineer roles starting May 2027.** Guitarist off the clock.

EDUCATION

New Jersey Institute of Technology **Sep 2025 - Present (Expected May 2027)**
Master of Science, Artificial Intelligence | GPA: 4.0/4.0 | Newark, NJ

Manipal Institute of Technology **Sep 2020 - Jun 2024**
Bachelor of Technology, Electronics and Instrumentation (Minor: Data Science) | CGPA: 8.08/10 | Manipal, India

TECHNICAL SKILLS

Languages: Python, C, C++, SQL, Cypher
Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, MONAI, YOLOv11, MiDaS, Vision Transformers, Diffusion Models, Recommender Systems, Computer Vision, LLM Agents, Model Context Protocol (MCP), Claude API
Data & Databases: Pandas, NumPy, Matplotlib, PySpark, MapReduce, MongoDB, Neo4j
Cloud & Tools: Azure AI Foundry, AWS EC2, FastAPI, Vercel, Render, Git, GitHub, CI/CD, Tableau, Retool, Mixpanel, Google Analytics 4

EXPERIENCE

Teaching Assistant **Sep 2026 - Present**
New Jersey Institute of Technology, Ying Wu College of Computing | Newark, NJ

- Support instruction for CS 106 (Introduction to Computing) and YWCC 207 (Computing and Effective Communication), leading office hours and lab sessions to help students debug code and build programming fundamentals.
- Evaluate programming assignments, technical reports, and presentations with structured feedback, and work with faculty to identify common student misconceptions and improve course materials.

Senior Executive, R&D **Nov 2024 - Jul 2025**
Montra Electric Tractor, TI Clean Mobility, Murugappa Group | Chennai, India

- Fine-tuned and optimized YOLOv11 (object detection) and MiDaS (monocular depth estimation) models for an autonomous fertilizer spraying system, reducing reliance on heavy sensor hardware and lowering system cost by 15-21%.
- Designed and tested the Battery Management System (BMS), cutting 0-80% charging time by 23 minutes, and integrated a 3 kW onboard charger enabling EV charging in low-grid regions without a transformer.

Machine Learning Engineer **May 2022 - Oct 2024**
Froker, Arroz Technologies | Bangalore, India

- Built and deployed an end-to-end recommendation engine for food, short videos, and prompts, increasing user engagement by 37% and orders by 11%.
- Built and optimized MongoDB data pipelines powering the recommendation engine and large-scale analysis, scaling internal analytics tooling into production infrastructure.
- Designed and deployed a multimodal content scoring pipeline using Azure AI Foundry APIs, evaluating multiple foundation models and tuning hyperparameters to meet platform evaluation requirements.
- Built Retool, Mixpanel, and Google Analytics 4 dashboards to drive data-driven product decisions and company-wide reporting.

Undergraduate Research Assistant, Embedded Systems **Sep 2021 - Feb 2022**
Manipal Institute of Technology | Manipal, India

- Built an autonomous line-following and obstacle-avoiding robot using closed-loop PID control over an IR sensor array and ultrasonic sensing for real-time obstacle detection and dynamic path rerouting.

PROJECTS

Trinetra | Autonomous Incident Response Multi-Agent System github.com/sairam782/Trinetra

- Built a multi-agent LLM system that monitors a live web application across alerts, logs, metrics, and traces, reasons over the full incident context, and selects an approved remediation runbook.
- Designed tiered agents (commander, telemetry, triage/adjudication, documentation) with a gated dry-run-first remediation executor, structured audit trail, and MCP-based tool integration.

GymCom | AI-Powered Live Workout Form Coach github.com/sairam782/GymCom

- Built a full-stack workout analysis app combining browser-side pose tracking, a FastAPI backend, and Claude-powered coaching reports, with real-time and uploaded-video modes.
- Implemented exercise-specific joint angle thresholds for rep counting, form scoring, and posture/symmetry feedback, with browser-side frame sampling to keep payloads lightweight on Vercel and Render.

Synthetic MRI Generation for Brain Tumor Classification

- Developed a 2D denoising diffusion model (MONAI) generating synthetic FLAIR MRI slices from BraTS-PED to address rare-tumor data scarcity, improving ResNet18 tumor detection accuracy from 82% to 88%.

Hybrid ResNet and Vision Transformer for Deepfake Detection

- Combined ResNet (local forgery artifacts) and ViT (global facial semantics) features, reaching up to 86% accuracy on Celeb-DF v2.